

# MATH 301: Homework 3

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## Sec 2.5 q2

*If  $S \subseteq \mathbb{R}, S \neq \emptyset$ , show that  $S$  is bounded IFF  $\exists I$ , a closed bounded interval s.t.  $S \subseteq I$*

Consider the forward and reverse statements to prove:

$$\begin{cases} S \text{ is a bounded set} \implies \exists I \text{ (a closed bounded interval) } \ni S \subseteq I \\ \exists I \text{ (a closed bounded interval) } \ni S \subseteq I \implies S \text{ is a bounded set} \end{cases}$$

Suppose  $S$  is a bounded set

$$\implies \exists a = \inf S \quad b = \sup S$$

$$\implies \forall s \in S, (s \leq b) \wedge (s \geq a)$$

$$\therefore S \subseteq [a, b] \equiv S \subseteq I$$

## Sec 2.5 q9

*If  $z \leq 0$  then  $z \notin K_1$ . If  $w > 0$  then it follows from the Archimedian Property that there exists  $n_w \in \mathbb{N}$  with  $w \notin (n_w, \infty) = K_{n_w}$*

## Sec 3.1 q5.a

*Use the definition of the limit of a sequence to establish the following limits:*

$$\lim_{n \rightarrow \infty} f(n) = 0, \quad f(n) := \left(\frac{n}{n^2 + 1}\right)$$

*Def (a.1): Limit Definition of Convergence*

$$\lim_{n \rightarrow \infty} f(n) = 0 \iff \forall \epsilon > 0, \exists N(\epsilon) \in \mathbb{N} \ni \forall n \geq N(\epsilon), |x_n - L| < \epsilon$$

### Backwards Search

| Step                                                             | Reasoning / Simplification                                         |
|------------------------------------------------------------------|--------------------------------------------------------------------|
| <hr/>                                                            |                                                                    |
| $\left  \frac{n}{n^2+1} - 0 \right  < \epsilon$                  | Limit definition statement                                         |
| $\left  \frac{n}{n^2+1} \right  < \epsilon$                      | Subtracting zero                                                   |
| $\frac{ n }{n^2+1} < \epsilon$                                   | Property of absolute values                                        |
| $\frac{n}{n^2+1} < \epsilon$                                     | Since $n \in \mathbb{N}$ , $n$ is always positive                  |
| $\frac{n}{n^2+1} < \frac{n}{n^2} = \frac{1}{n}$                  | Bounding: $n^2 + 1 > n^2 \implies \frac{1}{n^2+1} < \frac{1}{n^2}$ |
| $\frac{1}{n} < \epsilon$                                         | Sufficient condition for the inequality                            |
| $n > \frac{1}{\epsilon} \quad \frac{1}{\epsilon} \in \mathbb{R}$ |                                                                    |
| <hr/>                                                            |                                                                    |
| $\uparrow$                                                       | By Archimedian Property                                            |

Forward Logic: As  $\frac{1}{\epsilon} \in \mathbb{R}$ , this guarantees  $\exists N(\epsilon) \in \mathbb{N} \ni N(\epsilon) > \frac{1}{\epsilon}$ . This is the requirement to say  $\forall n \geq N(\epsilon), n > \frac{1}{\epsilon}$ . Therefore  $\frac{1}{n} < \epsilon$ . As  $\frac{n}{n^2+1}$  we can generalize the apply the inequality for  $f(n)$ . Thus we can say  $|f(n) - 0| < \epsilon$ . This satisfies the epsilon definition of a limit convergence of a sequence  $\therefore \lim_{n \rightarrow \infty} f(n) = 0$

## Sec 3.1 q5.d

*Use the definition of the limit of a sequence to establish the following limits:*

$$\lim_{n \rightarrow \infty} f(n) = \frac{1}{2}, \quad f(n) := \left( \frac{n^2 - 1}{2n^2 + 3} \right)$$

$$\lim_{n \rightarrow \infty} f(n) = 1/2 \iff \forall \epsilon > 0, \exists N(\epsilon) \in \mathbb{N} \ni \forall n \geq N(\epsilon), |f(n) - L| < \epsilon$$

### Backwards Search

| Step                                                            | Reasoning / Simplification                                         |
|-----------------------------------------------------------------|--------------------------------------------------------------------|
| $\left  \frac{n^2-1}{2n^2+3} - \frac{1}{2} \right  < \epsilon$  | Formal definition of a limit: $ f(n) - L  < \epsilon$              |
| $\left  \frac{2(n^2-1)-(2n^2+3)}{2(2n^2+3)} \right  < \epsilon$ | Find a common denominator: $2(2n^2 + 3)$                           |
| $\left  \frac{2n^2-2-2n^2-3}{4n^2+6} \right  < \epsilon$        | Distribute and simplify the numerator                              |
| $\left  \frac{-5}{4n^2+6} \right  < \epsilon$                   | Combine like terms                                                 |
| $\frac{5}{4n^2+6} < \epsilon$                                   | Apply absolute value: $ -5  = 5$ and $4n^2 + 6 > 0$                |
| $\frac{5}{4n^2+6} < \frac{5}{4n^2} < \epsilon$                  | Bounding: $4n^2 + 6 > 4n^2 \implies$ fraction is smaller           |
| $\frac{5}{4n^2} \leq \frac{5}{4n} < \epsilon$                   | Since $n \geq 1, n^2 \geq n$ , so $\frac{1}{n^2} \leq \frac{1}{n}$ |
| $\frac{5}{4n} < \epsilon$                                       | Sufficient condition to satisfy original inequality                |
| $\frac{1}{n} < \frac{4\epsilon}{5}$                             | Multiply by $\frac{4}{5}$ on both sides                            |
| $n > \frac{5}{4\epsilon}$                                       | Take the reciprocal (flips the inequality sign)                    |

↑ Choose  $N > \frac{5}{4\epsilon}$  by the Archimedean Property

Forward Logic: Let  $\epsilon > 0$  be given. As  $\frac{5}{4\epsilon} \in \mathbb{R}$ , the Archimedean Property guarantees there exists a natural number  $N(\epsilon) \in \mathbb{N}$  such that:  $N(\epsilon) > \frac{5}{4\epsilon}$ . This is the requirement to say that for all  $n \geq N(\epsilon)$ , the inequality  $n > \frac{5}{4\epsilon}$  holds. By taking the reciprocal and multiplying, we obtain:  $\frac{5}{4n} < \epsilon$ . Since  $n \geq 1$ , we know  $n^2 \geq n$ , which implies  $\frac{1}{n^2} \leq \frac{1}{n}$ . Thus, we can apply the following chain of inequalities to  $f(n)$ :  $\left| \frac{n^2-1}{2n^2+3} - \frac{1}{2} \right| = \frac{5}{4n^2+6} < \frac{5}{4n^2} \leq \frac{5}{4n}$ . Because we established  $\frac{5}{4n} < \epsilon$ , it follows by transitivity that:  $|f(n) - \frac{1}{2}| < \epsilon$ . This satisfies the epsilon definition of a limit for the convergence of a sequence.  $\therefore \lim_{n \rightarrow \infty} \frac{n^2-1}{2n^2+3} = \frac{1}{2}$

## Sec 3.1 q8

Prove that  $\lim(x_n) = 0 \iff \lim(|x_n|) = 0$ . Give an example to show that the convergence of  $(|x_n|)$  need not imply the convergence of  $(x_n)$ .

*Lemma (a.1) : Absolute Value Properties*

$$\forall a \in \mathbb{R}, |a| \implies \begin{cases} a \in \mathbb{P} \\ -a \in \mathbb{P} \\ a = 0 \end{cases}$$

$$\therefore |x_n| = (x_n \in \mathbb{P}) \oplus (-x_n \in \mathbb{P}) \oplus (x_n \in \{0\})$$

*Lemma (a.2) : Absolute Value Properties*

$$\text{Consider } ||x_n| - 0| = \begin{cases} (x_n \in \mathbb{P}) & \implies & ||x_n| - 0| = |x_n - 0| \\ (-x_n \in \mathbb{P}) & \implies & ||x_n| - 0| = | - (-x_n) - 0| = |x_n - 0| \\ (x_n \in \{0\}) & \implies & ||x_n| - 0| = |x_n - 0| \end{cases}$$

$$\therefore ||x_n| - 0| = |x_n - 0|$$

**Case  $(\rightarrow)$ :**  $\lim(x_n) = 0 \rightarrow \lim(|x_n|) = 0$

Observe  $||x_n| - 0| < \epsilon$  matches *Def (a.1)*. Naturally, should the inequality  $||x_n| - 0| = |x_n - 0|$  then this would imply their limits are the same as defined prior. This equality is given via *Lemma (a.2)*.

$$\therefore \lim(x_n) = 0 \rightarrow \lim(|x_n|) = 0$$

**Case  $(\leftarrow)$ :**  $\lim(x_n) = 0 \leftarrow \lim(|x_n|) = 0$

Applying the same *Def (a.1)*, *Lemma (a.2)* we arrive to the same conclusion following the same logic.

$$\therefore \lim(x_n) = 0 \leftarrow \lim(|x_n|) = 0$$

**Ex.**  $\lim(|x_n|)$  converges  $\nrightarrow \lim(x_n)$

*Lemma (a.3) :  $|(-1)^n| = 1$*

$$P(n) : |(-1)^n| = 1$$

$$P(1) : |-1| = 1$$

Suppose  $P(k) : |(-1)^k| = 1$  holds.

$$|(-1)^k| |(-1)| = 1 |(-1)|$$

$$P(k+1) : |(-1)^{k+1}| = 1$$

As  $P(k) \implies P(k+1) \therefore$  by PMI  $P(n)$  holds,  $\forall n \in \mathbb{N}$

For the sake of contradiction, suppose  $x_n := (-1)^n$  and  $\lim(|x_n|)$  converges  $\rightarrow \lim(x_n)$

$X := \{|(-1)^n| : n \in \mathbb{N}\} = \{1_1, 1_2, \dots, 1_n\}$  By *Lemma (a.3)*

The sequence never changes thus, is convergent on 1

Note:  $x_n = (-1)^n$  however is monotonic thus does not converge  $\nexists$

### Sec 3.1 q9

Show that  $\forall n \in \mathbb{N}, (x_n \geq 0) \wedge (\lim_{n \rightarrow \infty} x_n = 0) \implies \lim_{n \rightarrow \infty} \sqrt{x_n} = 0$

Suppose  $\forall n \in \mathbb{N}, (x_n \geq 0) \wedge (\lim_{n \rightarrow \infty} x_n = 0)$

$\lim_{n \rightarrow \infty} x_n = 0 \implies \exists n_\epsilon \text{ s.t. } \forall n \geq n_\epsilon, |x_n - 0| < \epsilon^2, (0 < \epsilon)$

$$0 < x_n < \epsilon^2 \iff 0 < \sqrt{x_n} < \epsilon$$

$\therefore \forall n \in \mathbb{N}, (x_n \geq 0) \wedge (\lim_{n \rightarrow \infty} x_n = 0) \implies \lim_{n \rightarrow \infty} \sqrt{x_n} = 0$

### Sec 3.1 q14

Prove:  $b \in \mathbb{R}, 0 < b < 1 \implies \lim_{n \rightarrow \infty} nb^n = 0$

*Theorem (a.1): Binomial Theorem*

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

*Expanded form:*

$$(a + b)^n = \binom{n}{0} a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \dots + \binom{n}{n} b^n$$

*Example for  $n = 3$ :*

$$(a + b)^3 = \binom{3}{0} a^3 + \binom{3}{1} a^2 b + \binom{3}{2} a b^2 + \binom{3}{3} b^3$$

$$= a^3 + 3a^2 b + 3a b^2 + b^3$$

$$b \in \mathbb{R}, 0 < b < 1 \implies \exists 0 < c \in \mathbb{R} \ni b = \frac{1}{1+c}$$

$$\implies b^n = \frac{1}{1+c^n}$$

$$\frac{1}{1+c^n} < \frac{n}{1+c^n}$$

$$\lim_{n \rightarrow \infty} \frac{n}{1+c^n} = 0 \implies \lim_{n \rightarrow \infty} \frac{1}{1+c^n} = 0$$

$$(1+c)^n = \sum_{k=1}^{\infty} \binom{n}{k} c^k \quad \text{By Theorem (a.1)}$$

**Sec 3.2 q2**

**Sec 3.2 q4**

**Sec 3.2 q6.b**

**Sec 3.2 q6.c**

**Sec 3.2 q9**